

Solar Technical Workshop

Gujarat Energy Research and Management Institute (GERMI) is organising a 'Comprehensive Course in Solar Photovoltaic Design, Technology and Application' for techno-management professionals who are actively involved in engineering, procurement and construction of solar photovoltaic power plants.

Course date: August 8-11, 2011

Course venue: Pandit Deendayal Petroleum University, Raisan Village, Gandhinagar, Gujarat

Key takeaways: The certification workshop will provide in-depth knowledge of fundamental PV principles, selection of components and designing of various types of PV systems, PV design and analysis software, hands-on building of PV systems, visit and study of 1 megawatt PV power plant.



Dr Omkar Jani (Left) with Prof. Brijesh Tripathi and Dr Manoj Kumar of the School of Solar Energy, Pandit Deendayal Petroleum University, at a test photovoltaic facility on campus

the huge target which has been set by the government, different companies are trying to engage themselves through either installation of power plants or design and manufacture of solar components, various devices and individual sub-systems required for the power plant. As far as the knowledge base is concerned, one is the engineering aspect of the power plant design and another is the physics of how this power gets generated. So the job opportunities exist in each of these sectors," opines P.K.Ghosh, management consultant, Agni Power & Electronics.

"As the megawatt-level solar power plants become more prominent. The employment opportunities of civil, mechanical and electrical engineering would go up considerably. Typically, a one megawatt plant during construction would need a minimum of five engineers in addition to technicians. The government itself has a target through the JNNSM of 1000 MW by 2013 and 20,000 MW by 2022. This offers numerous employment opportunities for engineers. To catch up with this, the manufacturing sector also would need engineers from chemical and mechanical disciplines," explains Jayakumar.

Subramanya adds, "The job outlook seems extremely positive and

enthusing. In fact, the solar industry currently suffers from severe shortage of skills at all levels for rapid growth and particularly in conceptualisation of opportunities, application engineering, marketing, financing, project management, O&M, etc."

"The two main prospects that I see in this industry are one on the manufacturing side and the other on the power projects side. There are auxiliary areas also such as electronics where things like inverters come into play. Now talking about the manufacturing side, we already do a lot of manufacturing of cells and modules in India but most of the raw materials and equipments are all imported. For example, for the module, most of the equipment and the machinery, solar cells (although solar cell manufacturing is now taking off in India) and backsheets polymers all come from outside. Similarly, when you talk about solar cell manufacturing, gases, metal pastes and silicon wafers are imported; so there is a lot of scope for starting manufacturing of these raw materials in India. And to cater to this, you would need engineers from various disciplines including electrical, electronics, physics, material science, mechanical and civil engineering,"

elaborates Dr Omkar Jani, principal research scientist, solar research wing at the Gujarat Energy Research and Management Institute—Research, Innovation and Incubation Centre (GERMI-RIIC).

He continues, "Now coming to the power projects side, you require engineers on the site for project execution—including engineers who can design the power plant at a higher level, as well as civil, structural, mechanical and electrical engineers for the project execution. There are also typical needs for businesses that can fabricate and supply mechanical structures like the Module Mounting Structures in a cost-effective way using cost-effective 'Indian-style' innovations. Thus, the industry opens up prospects at various levels as well. It's just like any other power industry. You need the same kind of people but then they have to be tuned for solar."

The manufacturing sector would need personnel with ITI, diploma and engineering qualifications. The implementation of projects would need diploma holders and engineers in addition to unskilled labour.

"A typical rule of thumb used in the US is that you need 40 people per installed megawatt of solar power, and these include people right from manufacturing to installation to maintenance. But if you look at the Indian scenario, this number needs to be scaled up to 100 because we typically employ more manpower for manufacturing and execution of projects. So now you can extrapolate how many megawatts is India looking to install. As per the objectives of the JNNSM, the government plans to increase its deployment to 20,000 MW by 2022, which clearly points you to the kind of growth we are looking at in solar industry," explains Dr Jani.

What role to choose?

In solar industry, you can find a place in any of the following broad categories: design engineers, who take care of the design and engineering of systems; production engineers, who oversee